

Small Mammal Dermatology

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Lecture Overview

At UC Davis, almost one-third of all rabbits brought to the Veterinary Medical Teaching Hospital had evidence of skin disease and almost 50% of guinea pigs, highlighting the importance of dermatology in this group of “special” companion animals. In this 50-minute lecture we will review the most common dermatologic presentations in the most common species of exotic companion mammals presented in private practice. If you are interested in a deeper dive into additional dermatologic conditions, consider:

- relevant chapters in 3rd Edition of [Ferrets, Rabbits and Rodents Clinical Medicine and Surgery](#)
- chapter 2 of [Current Therapy of Exotic Pet Practice](#).

Lecture Outcomes

- Identify normal glandular and dermatologic anatomy features in common zoological companion mammal species.
- Compare the expected signalment, presentation, diagnostics and therapeutics recommended for common infectious, endocrine and neoplastic dermatopathies reported in different ZCM species.
- Distinguish which antiparasitic(s) are contraindicated in which ZCM species.

Lecture Notes

Section 1: Different species have different normal dermatologic anatomy!

So many species, so little time! There are several common species that may present for veterinary care in companion animal practice. Each of these species has their own unique dermatological anatomy.

Many small mammals have glands that are involved in scent marking their environment. Pay attention to which species has which glands, and which ones are often clinically associated with pathology!

- Rabbits have chin glands and may be seen “chinning” items to mark them with their scent. The glands are rarely enlarged and are not able to be palpated or directly observed. The bilateral inguinal scent glands of rabbits are readily visualized on either side of the anal opening. A waxy, brown, malodorous semi solid excretion is created here and are used for marking. Normal, healthy rabbits are able to clean their scent glands on their own, however, those suffering from back pain, paraplegia – paralysis, or generally weak may not be able to clean their own scent glands, requiring aid from their human family members to keep them clean (there are lots of online resources for clients to learn how to do this, but here is [one from a trusted source](#)...). This site is often a site affected by myiasis or fly strike.
- Guinea pigs have a single “grease” gland, also called the supracaudal gland. Placed directly on midline on the rump, this area may be waxy or greasy, particularly in intact males. Most guinea will be able to keep this site clean, but older or sick guinea pigs may need aid from their human family members to help them (don’t worry, [there is a trusted source](#) for this activity as well!). Outside of being a bit impacted with waxy material; this site is not typically a site of pathology.

- Ferrets, like all mustelids, have anal glands. However, ferrets bred at Marshall's Farms in North America, identified based upon two tattoos in the right ear, are spayed or neutered and descented (anal saccullectomy) at 4-6 weeks of age. The distinctive ferret odor is caused by diffusely distributed sebaceous glands that are more active (thus, more odiferous) in sexually intact ferrets.
- Syrian or golden hamsters have bilateral darkly pigmented circular to elliptical flank glands that are more pronounced in males as they are under testosterone influence. Although an uncommon site for pathology, I have seen melanomas arise from these structures.
- Dwarf hamsters (Campbell's, Roborovski, Russian/Siberian/winter/stripped) have a single ventral midline sebaceous gland that is larger in males. In males it is often raised, slightly crusty and brown in color, while it may be no more than a slightly raised dot in females. This is an uncommon site for neoplasia in the species.
- Mongolian gerbils also have a midline ventral gland, like dwarf hamsters in appearance. This is a common site for neoplasia in the species.

A few of the non-traditional mammals have some additional unique dermatologic structures you should be aware of.

- Chinchillas have a super dense pelage with more than 60 hairs per follicle, to aid in thermoregulation for this high-altitude species. As a defense mechanism to escape predators, chinchillas can drop chunks of this dense fur, called "fur slip" leaving behind a smooth, non-irritated patch of skin. If restrained inappropriately, this is a common sequela.
- Hedgehogs have furred skin along their ventrum and sides and deeply muscular mantle that is ornamented with keratinized spines. Although "Prince Quilliam" is a deeply amusing name, hedgehogs DO NOT HAVE QUILLS. Quills are a dermatologic feature of porcupines and a few other species and easily epilate for defense. Their often-barbed nature makes them challenging to remove from tissues that they embed in. In comparison, the non-barbed spines of hedgehogs are semipermanent and only exfoliate during an annual "molt" or shed, that is called "quilling" (not confusing at all!).

Section 2: The most commonly reported dermatologic diseases differs by species.

2.1 Acariasis

Acariasis (mite infestations) are common in exotic companion (small) mammals. Several different classifications of mites affect these species including fur mites, burrowing skin mites, and ear mites. While the presentation for each of these diagnoses may mirror the clinical signs seen in dogs and cats, there are some distinct clinical presentations that you should recognize that are unique to small mammals.

Fur mites most commonly affect rabbits (*Cheyletiella parasitovorax*) and less so in rats (*Radfordia ensifera*), mice (*Myocoptes musculus*) and guinea pigs (*Chirodiscoides caviae*). As these mites ingest shed epidermal cells, they don't always cause clinical pruritic except in more severe cases. The most common clinical finding is scaling and dandruff, classically if observed closely the dandruff may be seen to be moving, thus the colloquialism "walking dandruff". Distribution of dandruff in rabbits is classically over the dorsal cervical and saddle region while more diffuse scale and dandruff is noted in rats. Confirmation of the diagnosis is made through a tape prep of the dandruff, and detection of mites microscopically. *Cheyletiella* enjoys the fur of most species and is not above hopping onto humans (zoonotic) or other furred animals in the home (cats, dogs, guinea pigs, etc....)

Skin mites are often diagnosed in hedgehogs (*Caparinia* spp. *Chorioptes* spp, *Notoedres* spp.), guinea pigs (*Trixacarus caviae*, *Sarcoptes scabiei*), rats (*Ornithonyssus bacoti*) and Syrian hamsters (*Demodex* spp.). Clinical signs and presentation mirror that seen in dogs and cats with pruritic, crusting, alopecia and barbering of hairs. Some species-specific presentations you should be aware of include self-mutilation, that is most common in hedgehogs, who may rub their mantle along cage bars creating a wound or bite and chew at their feet, sometimes removing digits or parts of limbs. A classic presentation of acariasis in young guinea pigs is seizure activity, which is postulated to be secondary to the extreme pruritis. At times the neurologic events precede clinically relevant skin lesions. Diagnosis of these mites requires a skin scrape and microscopic evaluation of the mites except in rats, where they can often be seen as yellow-brown-orange moving structures within the hairs. While most of the mites are species specific, the tropical rat mite (*O. bacoti*) is not that picky and will infest humans and other mammals in the environment.

Ear mites are a common diagnosis in ferrets (*Otodectes cynotis*) and rabbits (*Psoroptes cuniculi*). Infestations in both species may cause aural pruritis and discharge. The classic appearance of ear mites in ferrets is a “coffee grounds” appearance to the aural discharge. Rabbits suffering from ear mites have scabbed ulcerated lesions within their ear canal and are extremely painful. This should be considered before collecting samples for diagnosis and minimal to no cleaning of the canals should be performed until treatment is started. Diagnosis is relatively straight forward, with collection of aural discharge in mineral oil examined microscopically. Since most ferrets are subclinically infected and *Otodectes* can infect other carnivores in the home, screening of ferrets during wellness visits is warranted.

Treatment of acariasis is based upon the host species being treated and, in some regards, the type of mite being treated for. Some important considerations for treatment:

- Choosing a miticide. The lowest bar of which antiparasitic agent to choose should be safety. Fipronil is toxic to rabbits and contraindicated for use in this species. Exposed rabbits develop neurologic signs and there are no reports of survival after developing toxicity. There are anecdotal reports of fipronil toxicity in hedgehogs, and the product should be used with caution in these species. Outside of these toxic considerations, multiple resources should be utilized to choose an effective and safe antiparasitic for the host species and diagnosis including textbook chapters, literature search, and formularies. In a pinch, selamectin (Revolution) has been deemed safe in most species, although the dose (in mg/kg) is much higher than the typical doses recommended for dogs and cats.
- Treating the environment. Some mites may live for an extended period off the host without a blood meal (Ex: *Ornithonyssus bacoti*), requiring environmental cleaning during the treatment period (vacuuming, laundering, high temperature washing of food dishes).
- Treat all infested (or potentially infested) hosts in the household. There are some mites that are indiscriminate feeders, requiring treating all mammals in the household to clear the infection (ex: *Otodectes cynotis*, *Cheyletiella parasitovorax*)

2.2 Dermatophytosis

Everyone loves fun with fungus! “Ringworm” or dermatophytosis is commonly reported in guinea pigs, hedgehogs and chinchillas. Typically, patients are young, recently obtained or otherwise immunocompromised. *Trichophyton* spp. are the most identified isolates in these species. Like dogs and cats, patients will present with a crusting dermatopathy that exhibits variable pruritis. Classically, guinea pigs get lesions on their face and muzzle and chinchillas have lesions centered around their feet and legs. The same in office diagnostics that would be pursued to detect fungi for dogs and cats can be pursued for small mammals including cytology and trichograms. *Trichophyton* spp. does not typically fluoresce

with Wood's lamp illumination, making it a less useful diagnostic. Cultures and fungal specific PCRs can also be pursued.

Therapeutic intervention should focus on systemic antifungal use, as topical-only treatment regimens are unlikely to be successful. Doses for itraconazole and terbinafine are available in formularies, textbooks and the literature. Itraconazole toxicity is not common in these species.

2.3 Neoplasia

Ferrets commonly present with mast cell tumors that can be present on any dermatologic site on the body, although they are commonly noted over the trunk. In comparison to other species, the biological behavior of this tumor is benign, with rare instances of metastasis. The classic clinical appearance is a raised, red, and ulcerohemorrhagic lesion, the lack of this distinctive appearance should not preclude clinical suspicion. Clients often describe pruritis and a "waxing and waning" appearance to the mass. The diagnosis of mast cell tumor can be made through cytologic evaluation of an aspirate or touch impression from an ulcerative lesion, showcasing round cell neoplasm with abundant purple granules; biopsy is rarely required for diagnosis. Given the benign biological behavior, masses can be managed with topical products to reduce pruritis (diphenhydramine and/or steroid preparations). Surgical management is often reserved for masses that interfere with normal activity or cause excessive discomfort (periocular arising masses).

Guinea pigs commonly present with trichofolliculomas, benign masses that classically arise from the dorsolateral rump region in boars (male guinea pigs). Despite their benign nature, these masses can become quite large, become ulcerated and secondarily infected, and in some cases, impede normal ambulation. Cytology can be very helpful in assessing if the mass is secondarily infected, however, neoplastic cells are rarely found in aspirate preps. The most prominent cytologic feature is abundant keratin debris and confirmation of diagnosis often requires histopathology. Surgery with complete margins is curative but can be complicated in cases with very large masses.

A dermatologic paraneoplastic syndrome of rabbits is associated with the diagnosis of thymoma. Clinical signs include a progressive non pruritic, scaly, flaky, dermatitis. Distribution often starts around the head and shoulders but often progresses to a more diffuse pattern. Diagnosis can be challenging, as often skin-specific diagnostics looking for infectious etiologies are pursued to rule out dermatophytosis and ectoparasites, yielding negative results. A skin biopsy in these cases may reveal the loss of sebaceous glands and a preliminary diagnosis of sebaceous adenitis, a poorly understood immune mediated dermatopathy in rabbits. A thoracic radiograph should be performed to rule out presence of a cranial mediastinal mass. Rabbits with thymoma may have their dermatologic clinical signs improve under radiation treatment for thymoma, but only minimally improve at best when treated medically.

2.4 Endocrine dermatopathies

Bilateral truncal/tail alopecia in a ferret is the most common clinical presentation of adrenocortical disease. In this species, adrenocortical hypertrophy leads to androgen hormone elevations, thus additional clinical signs may include an enlarged or swollen vulva in females or stranguria/dysuria in males. Interestingly, up to 30% of ferrets with adrenocortical disease exhibit pruritis, that is sometimes intense, complicating the initial differential diagnosis list as more classically pruritic diseases are often prioritized. Either abdominal ultrasound imaging and/or serum androgen concentrations can be used to confirm the diagnosis. Adrenocortical disease is medically managed with intermittent (typically annually) placed deslorelin implants, a GnRH super agonist or surgically managed with adrenalectomy.

Bilateral truncal alopecia in a female guinea pig is a dermatologic sequela in guinea pigs with hormone-producing paraovarian cysts. While many intact sows will have paraovarian cysts, only a minority of those cysts will produce hormones. The alopecia is progressive, non-pruritic, and the skin lacks any gross pathology. A mid-abdominal mass may or may not be palpable. Ultrasound evaluation at the level of the paralumbar fossa can aid in identification of the cysts. Hair regrowth is possible after surgical management (ovariectomy) or medical management with hormone therapy.